Retail Sales Performance & Customer Insights

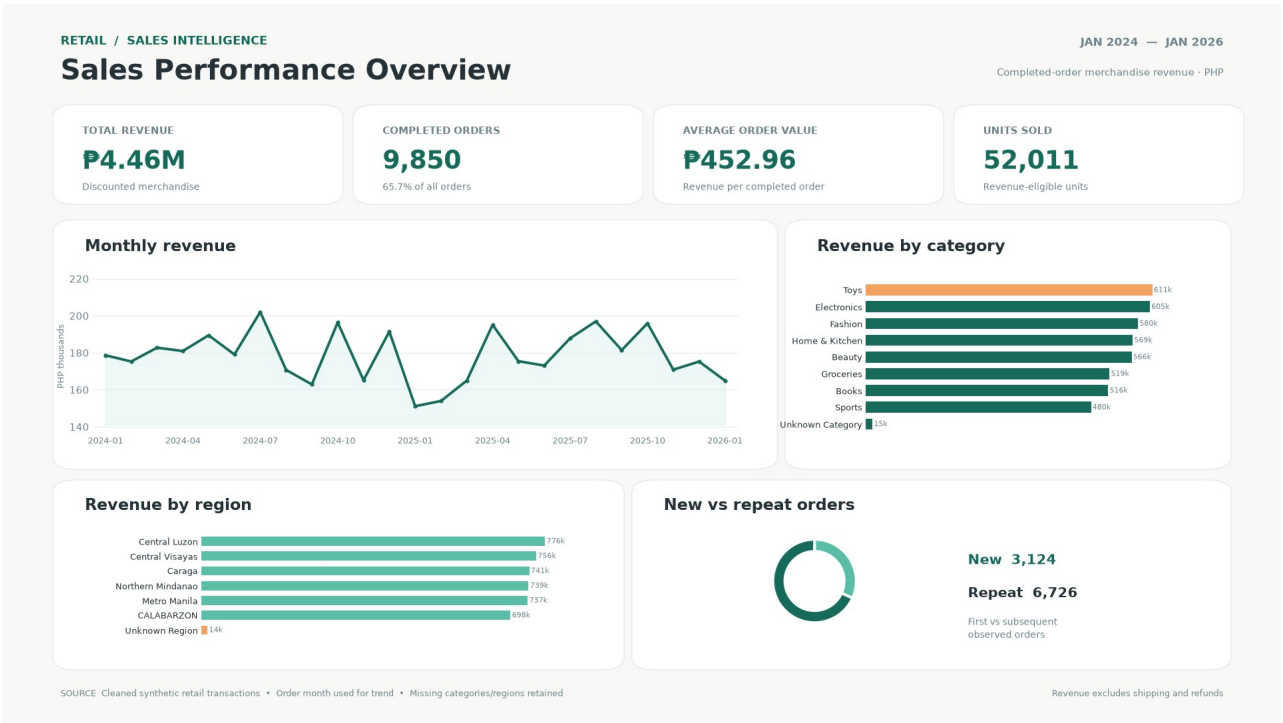
Option A | Sales Performance & Customer Insights

Synthetic Philippine retail transactions • January 2024–January 2026

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Submission date: November 7, 2026

Technology: Python • pandas • Matplotlib • Jupyter Notebook



Executive dashboard preview (full-size figure reproduced in Section 8).

1. Executive Summary

This project evaluates a synthetic Philippine retail business using seven linked CSV datasets and a reproducible Python workflow. The objective is to quantify completed-order merchandise sales, identify major revenue contributors, describe customer purchasing behavior, and examine cancellations and returns. Data were profiled for missing values, inconsistent categories, duplicate identifiers, mixed formats, and invalid references before building an order-item-level analytical fact table.

Of 15,000 orders, 9,850 were Completed (65.67%), producing PHP 4,461,673.07 in discounted merchandise revenue and an average order value of PHP 452.96. Annual revenue decreased 2.44% from 2024 to 2025, alongside a decline in completed orders. Toys and Electronics were the highest-revenue categories; Central Luzon led the identified regions. The top ten products represented only 4.14% of revenue, indicating limited concentration among leading products. Among completed orders, 3,124 were first observed and 6,726 subsequent orders, under a dataset-window definition. Cancelled and Returned statuses accounted for 8.23% and 3.85% of all orders, respectively.

Recommended next steps are a monthly KPI monitoring framework and structured investigation of cancellation and return reasons. Results are descriptive, not causal, and synthetic data cannot establish real-world Philippine retail demand.

2. Project Overview

Business question: How is the store performing, and what drives revenue? The project covers data loading, profiling, cleaning, fact-table integration, KPI analysis, five charts, an executive dashboard, and recommendations. The reporting period is January 2024 through January 2026; the final year contains only January.

3. Dataset Overview and Grain

Source table	Rows	Role
customers	5,000	Customer attributes / geography
products	800	Product, category, list price
orders	15,000	Order date, status, customer, discounts
order_items	37,501	Line-item quantity and purchase price
payments	15,000	Payment reference and amount
shipments	15,000	Fulfillment dates and shipping
returns	1,200	Return and refund records

The analytical sales fact table is at one row per order item, joining order_items to orders, products, and customers. Distinct order IDs are used for order-based KPIs to prevent counting each line item as a separate order. Payment, shipment, and return tables were assessed separately rather than multiplying fact rows through inappropriate joins.

4. Data Cleaning and Validation Log

Issue / check	Observed evidence	Treatment and rationale
Category inconsistencies	18 variants across products	Mapped to 8 standardized categories; 800 products retained.
Missing order discounts	756 orders (5.04%)	Used observed median 0.07 for analysis; retained missing-value

		flag and sensitivity discussion.
Missing item prices	2,236 item rows	Imputed 2,227 from matched product list prices; 9 unresolved orphan-product rows excluded from revenue eligibility.
Currency formats	3,229 currency-formatted values	Removed currency symbols/separators and parsed numeric values.
Zero quantities	372 item rows	Flagged and excluded from revenue-eligible merchandise calculations.
Invalid customer references	43 orders	Preserved with unknown geography; did not silently drop transactions.
Invalid product references	129 item rows	Preserved and flagged; unresolved category/price treated explicitly.
Refund contradictions	280 of 1,200 return records	Flagged, not automatically overwritten; refund totals excluded from defined revenue.
Delivery dates	729 missing	Documented missingness; did not fabricate delivery dates.
Duplicate and key checks	Unique primary keys after validation	Exact-row duplicates and duplicate key definitions checked; no duplicate primary keys remained in four core tables.

Additional missingness was documented in customer age, email, phone and loyalty tier; product stock; order shipping fee; payment amount; and return status/amount. Missing values were not indiscriminately filled. Cleaning decisions distinguish analytical imputations, retained unknowns, and records excluded only from a specific calculation.

5. Modeling and Metric Methodology

Sales revenue = quantity × unit price at purchase × (1 − order discount), counted only for Completed orders with positive quantity and non-missing valid price. Shipping fees and refunds are excluded. Completed orders = distinct order_id with Completed status. Average order value = completed-order merchandise revenue / completed orders. Revenue by month uses original order month; category and region use the joined product/customer attributes, retaining missing groups for reconciliation.

First observed ("New") versus Repeat completed orders: sort each customer's completed orders by date and order identifier; classify the first observed completed order as New and subsequent completed orders as Repeat. This is not a lifetime acquisition or retention measure. Order cancellation and return rates use all orders and recorded order statuses; month is original order month, not cancellation/return event month.

Validated KPI	Result
Completed-order merchandise revenue	PHP 4,461,673.07
Total orders / Completed orders	15,000 / 9,850
Completed-order share	65.67%
Average order value	PHP 452.96
Revenue-eligible units	52,011
Revenue-eligible line items	24,404
2024 / 2025 revenue	PHP 2,174,844.01 / PHP 2,121,862.45
New / repeat completed orders	3,124 / 6,726
Cancelled / Returned orders	1,235 / 578

Quality assurance: the notebook passed 15 automated checks covering order counts, revenue reconciliation, figure exports, category totals/units, and new/repeat counts and completed revenue. The dashboard image was separately confirmed at 1920 × 1080 pixels.

6. Revenue and Customer Analysis

Category revenue: Toys PHP 610,562.72 (13.68%); Electronics PHP 605,362.95 (13.57%); Fashion PHP 580,005.88 (13.00%). No identified category exceeded 14%. Unclassified category revenue of PHP 14,675.96 (0.33%) was retained. Category totals reconcile to the overall revenue KPI.

Geography: Central Luzon generated PHP 776,432.08 (17.40%), ahead of Central Visayas at PHP 756,340.57. Unknown-region revenue totaled PHP 14,299.98 and is associated with invalid customer ID CUST999999. This issue is surfaced rather than discarded.

Customer spending: 4,307 customers contributed positive completed-order revenue; mean PHP 1,035.91 and median PHP 881.91. The right-skewed distribution has a small high-spending tail. The spending histogram displays values through the 99th percentile for readability; summary statistics retain all 4,307 customers.

7. Required Visualizations

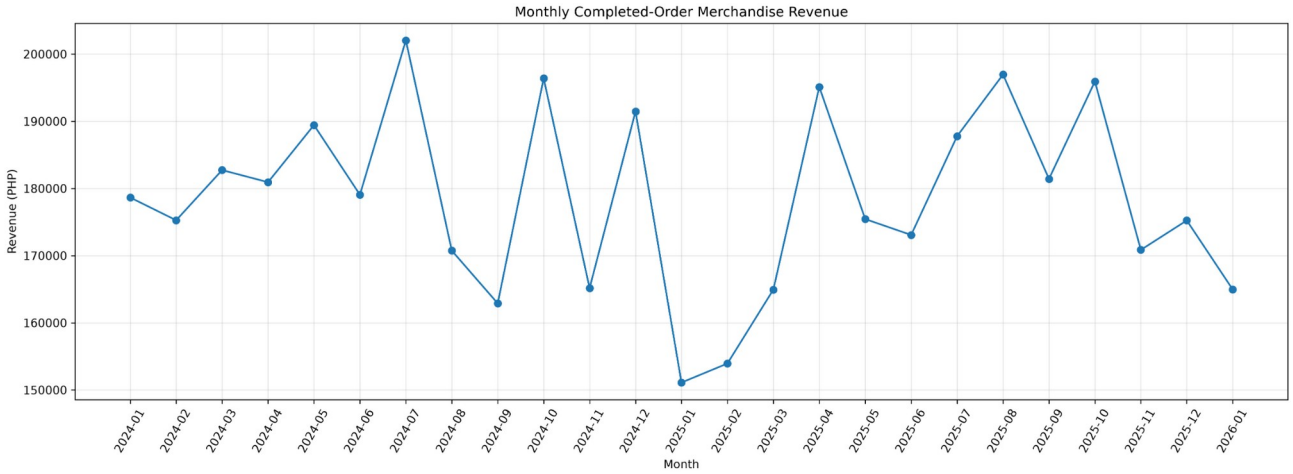


Figure 1. Monthly completed-order merchandise revenue; 2026 includes January only.

Revenue varies by month, with 2025 annual revenue 2.44% below 2024; no stable monotonic trend is established.

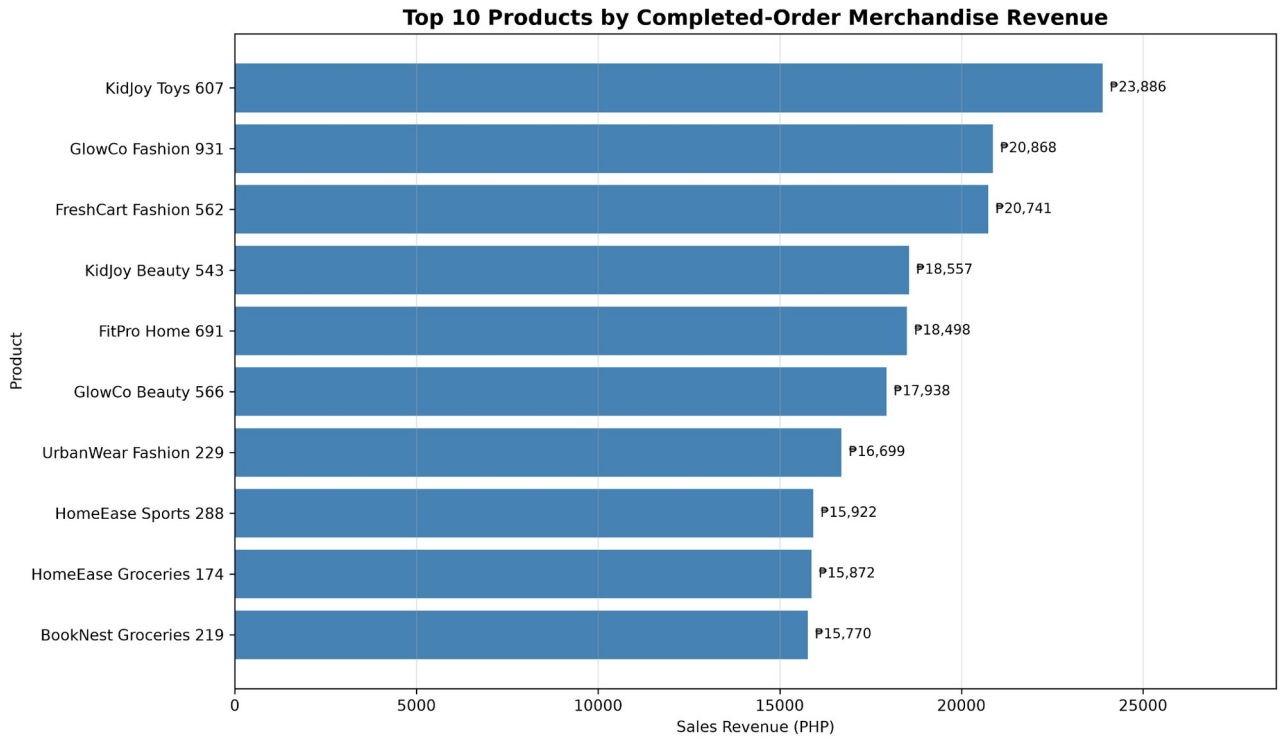


Figure 2. Top ten products by completed-order merchandise revenue.

The top ten products together account for PHP 184,750.83 (4.14%) of revenue; KidJoy Toys 607 leads at PHP 23,886.20.

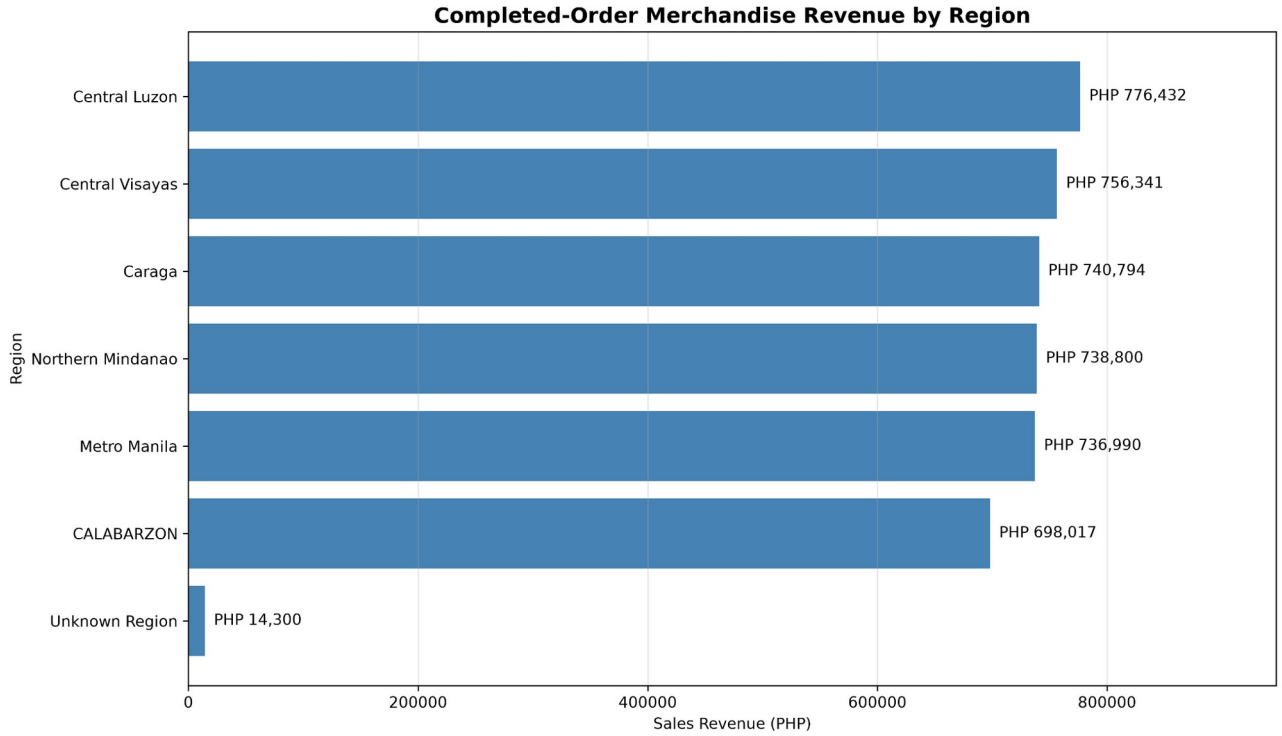


Figure 3. Revenue by customer region, including unknown region.

Central Luzon leads identified regions, while unknown-region transactions remain visible for data-quality transparency.

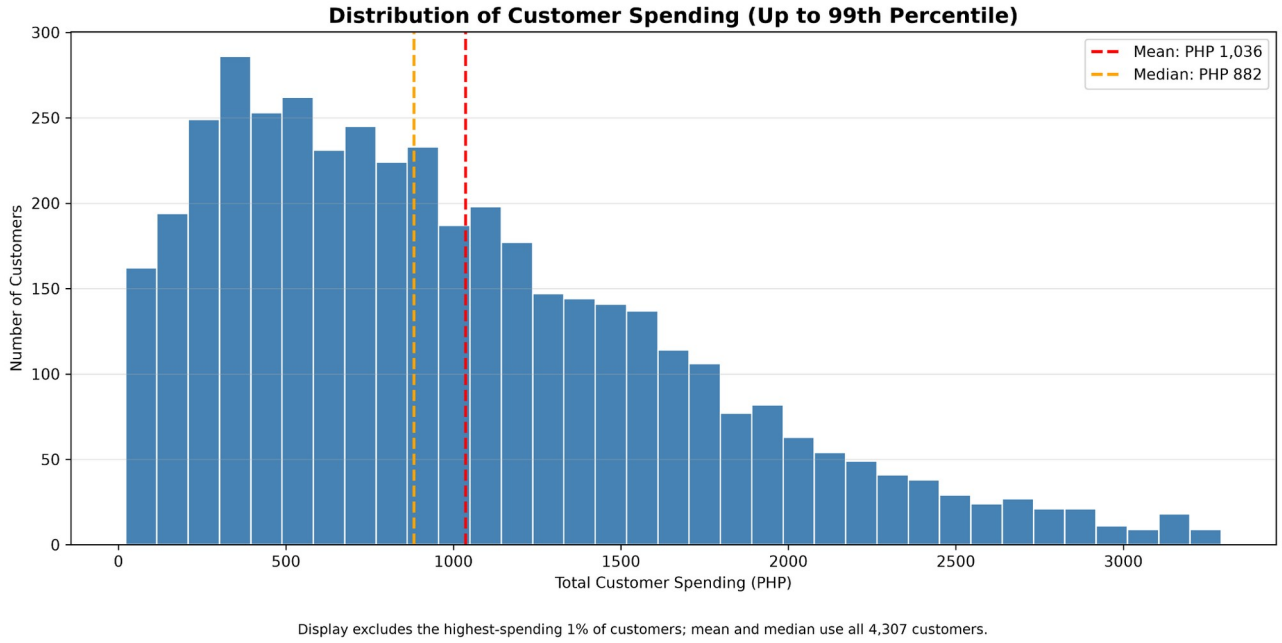


Figure 4. Customer spending distribution, displayed to the 99th percentile.

Mean spending exceeds median spending, consistent with a positively skewed distribution.

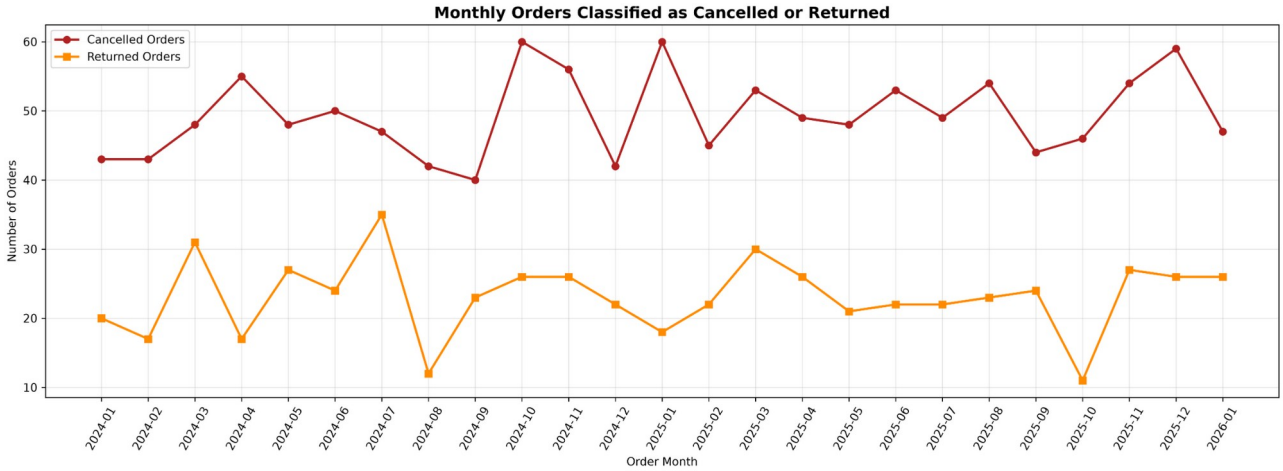


Figure 5. Monthly order counts classified as Cancelled or Returned.

Cancellation and return series are based on recorded status and original order month; they are not event-date timelines.

8. Executive Dashboard — 1920 × 1080

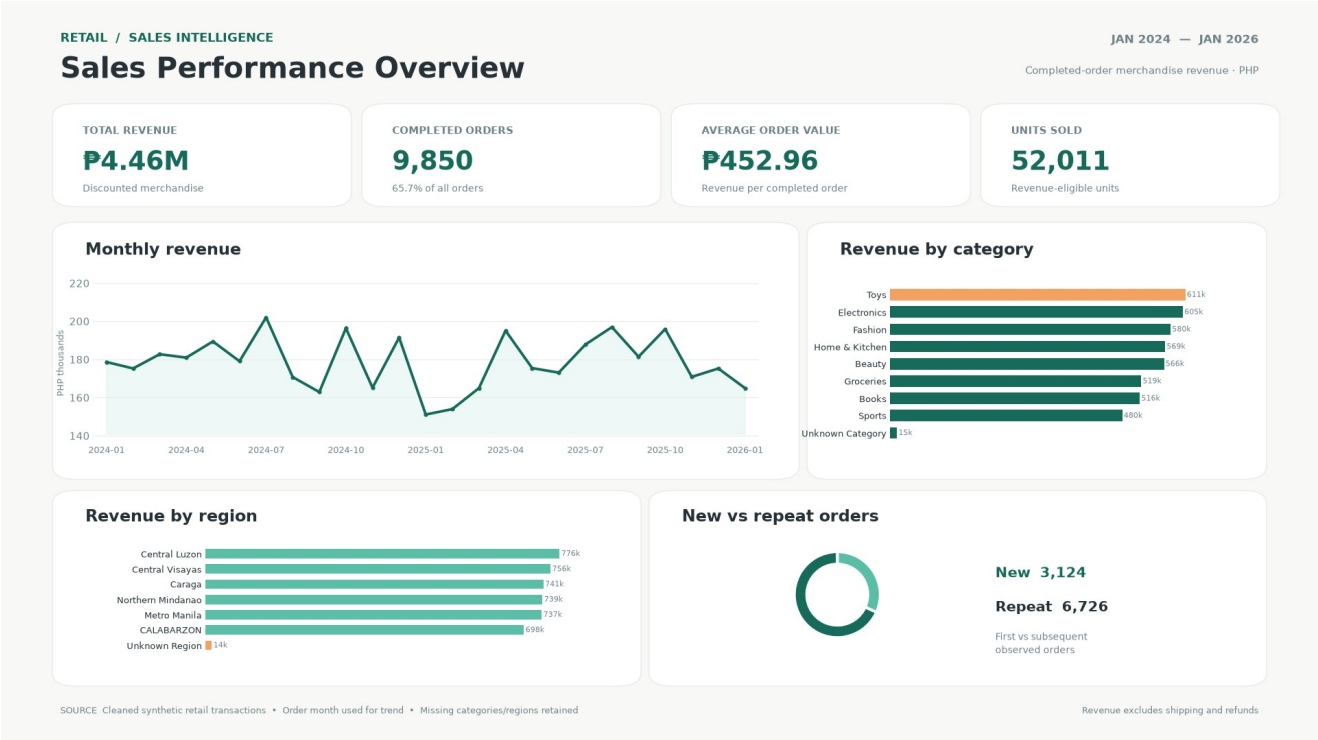


Figure 6. Forest Green & Cream executive overview, exported at exactly 1920 × 1080 pixels.

The dashboard prioritizes revenue, completed orders, average order value and revenue-eligible units, then connects trend, category, region and first-versus-subsequent completed-order views. The five detailed chart PNGs above remain the primary rubric visualization evidence.

9. Key Takeaways and Insights

- Completed-order merchandise revenue was PHP 4.46M from 9,850 completed orders; 65.67% of all orders were Completed.
- 2025 revenue declined 2.44% versus 2024, alongside fewer completed orders (4,661 versus 4,817).
- Toys (13.68%) and Electronics (13.57%) were leading categories, but no identified category exceeded 14% of revenue.
- The top ten products generated only 4.14% of total revenue; product-level performance is not dominated by a few bestsellers.
- Central Luzon contributed PHP 776K (17.40%), with relatively similar totals across the six identified regions.
- Customer spending was right-skewed (mean PHP 1,036; median PHP 882), supporting further segmentation by frequency and basket size.
- Of 9,850 completed orders, 6,726 (68.3%) were subsequent observed purchases, not a measured lifetime retention rate.
- Cancelled and Returned statuses represented 8.23% and 3.85% of all orders; 280 return records had potentially inconsistent refund status/amount combinations.

10. Business Recommendations

Recommendation 1 — Monthly sales performance monitoring

Implement a recurring scorecard tracking completed-order revenue, completed orders, AOV, year-over-year and month-over-month comparisons, category mix and region mix. Review volume and AOV together before attributing revenue changes to customer demand. Suggested measures: revenue growth, completed-order growth, AOV and reporting cadence.

Recommendation 2 — Investigate cancellation and return drivers

Capture structured reason codes and analyze cancellations/returns by product, category, region, order value and fulfillment stage. Track monthly rates with denominators and avoid treating original order month as event date. Suggested measures: cancellation rate, return rate, and percentage of cases with documented reasons.

11. Conclusion and Limitations

The project demonstrates a reproducible workflow from source profiling through cleaning, fact-table modeling, validated KPIs, visualization and business interpretation. Its main contribution is a traceable performance baseline rather than causal explanation. Findings are limited by synthetic source data, incomplete 2026 coverage, unresolved reference and refund anomalies, the completed-order merchandise revenue definition, and absence of cost/margin or causal drivers. Customer “new” status refers only to first observed completed order within the dataset.

Appendix — Submission and Reproducibility Notes

Notebook: retail_sales_analysis.ipynb (validated capstone analysis). Dashboard generator: retail_sales_dashboard_forest_green.ipynb. Cleaned datasets: seven *_cleaned.csv files in data/cleaned/. Figures: five detailed PNGs and one 1920 × 1080 dashboard PNG in outputs/figures/. Recommended repository: README, notebook(s), cleaned customer/product CSVs, figures, one-page insights summary, and this PDF. Source notebooks contain the detailed code, cleaning investigations, validation outputs, and chart-generation evidence.

Selected code-and-output excerpts rendered directly from the saved Jupyter notebook (.ipynb). These are representative evidence, not screenshots of every cell. Full executable code and outputs remain available in the notebook.

01 / Cleaning: missing purchase prices

Source: analysis notebook, cell 80 (zero-based index 79); saved code and output.

01 / Cleaning: missing purchase prices

In [ ]: Code excerpt from saved notebook

```
# Impute missing purchase prices using validated catalog prices

order_items = cleaned_datasets["order_items"]

# Preserve information about which prices were originally missing
order_items["price_imputed"] = False

# Create a lookup from product ID to catalog price
catalog_prices = products.set_index("product_id")["unit_price"]

# Identify missing purchase prices
missing_mask = order_items["unit_price_at_purchase"].isna()

# Look up the corresponding catalog price
replacement_prices = order_items["product_id"].map(catalog_prices)

# Identify records eligible for imputation
impute_mask = missing_mask & replacement_prices.notna()

# Fill only eligible missing prices
order_items.loc[
    impute_mask,
```

Saved output:

```
Purchase prices imputed: 2227
Remaining missing purchase prices: 9
Total order items preserved: 37501
```

02 / Modeling: sales fact table

Source: analysis notebook, cell 155 (zero-based index 154); saved code and output.

02 / Modeling: sales fact table

In [ ]: Code excerpt from saved notebook

```
# Construct initial sales fact table

sales_fact = (
    order_items
    .merge(
        orders,
        on="order_id",
        how="left",
        validate="many_to_one"
    )
    .merge(
        products,
        on="product_id",
        how="left",
        validate="many_to_one",
        suffixes=("", "_product")
    )
    .merge(
        customers,
        on="customer_id",
        how="left",
        validate="many_to_one",
        suffixes=("", "_customer")
    )
)
```

Saved output:

```
Original order item rows: 37501
Sales fact table rows: 37501

Unique order item IDs:
37501

Sales fact table shape:
(37501, 36)
```

03 / KPI calculation and output

Source: analysis notebook, cell 166 (zero-based index 165); saved code and output.

03 / KPI calculation and output

In [ ]: Code excerpt from saved notebook

```
# Calculate core sales performance KPIs

total_revenue = sales_fact["sales_revenue"].sum()

total_completed_orders = (
    orders.loc[
        orders["order_status"] == "Completed",
        "order_id"
    ].nunique()
)

average_order_value = (
    total_revenue / total_completed_orders
    if total_completed_orders > 0 else 0
)

total_units_sold = (
    sales_fact.loc[
        sales_fact["revenue_eligible"],
        "quantity"
    ].sum()
)

print("SALES PERFORMANCE KPIs")
print("-" * 40)

print(f"Total Sales Revenue: ₱{total_revenue:,.2f}")
print(f"Total Completed Orders: {total_completed_orders:,}")
print(f"Average Order Value: ₱{average_order_value:,.2f}")
print(f"Total Units Sold: {total_units_sold:,}")
```

Saved output:

```
SALES PERFORMANCE KPIs
----------------------------------------
Total Sales Revenue: ₱4,461,673.07
Total Completed Orders: 9,850
Average Order Value: ₱452.96
Total Units Sold: 52,011
```

04 / Final validation results

Source: analysis notebook, cell 213 (zero-based index 212); saved code and output.

04 / Final validation results

In [ ]: Code excerpt from saved notebook

```
from pathlib import Path

expected = {
    "Order count": (len(orders), 15000),
    "Completed orders": (int(orders["order_status"].eq("Completed").sum()), 9850),
    "Revenue-eligible rows": (int(sales_fact["revenue_eligible"].sum()), 24404),
    "Cancelled orders": (int(orders["order_status"].eq("Cancelled").sum()), 1235),
    "Returned orders": (int(orders["order_status"].eq("Returned").sum()), 578),
}
results = {}
for label, (actual, target) in expected.items():
    results[label] = actual == target
    print(f'{"PASS" if results[label] else "FAIL"} | {label}: {actual:,} (expected {target:,})')

actual_revenue = float(sales_fact["sales_revenue"].sum())
results["Revenue reconciliation"] = abs(actual_revenue - 4461673.07) < 0.01
print(f'{"PASS" if results["Revenue reconciliation"] else "FAIL"} | Revenue: PHP
{actual_revenue:,.2f}')

figures = [
    "monthly_revenue_trend.png", "top_10_products_revenue.png",
    "regional_revenue.png", "customer_spending_distribution.png",
    "monthly_cancellations_returns.png",
]
```

Saved output:

```
PASS | Order count: 15,000 (expected 15,000)
PASS | Completed orders: 9,850 (expected 9,850)
PASS | Revenue-eligible rows: 24,404 (expected 24,404)
PASS | Cancelled orders: 1,235 (expected 1,235)
PASS | Returned orders: 578 (expected 578)
PASS | Revenue: PHP 4,461,673.07
PASS | monthly_revenue_trend.png
PASS | top_10_products_revenue.png
PASS | regional_revenue.png
PASS | customer_spending_distribution.png
PASS | monthly_cancellations_returns.png
PASS | Category revenue
PASS | Category units
PASS | New/repeat order counts
PASS | New/repeat completed revenue

ALL CHECKS PASSED
```